© 2021 ASHRAE, Inc.

SAND POINT, AK, USA

WMO: **703165**

Lat: **55.319N** Lon: **160.521W** Elev: **6** StdP: **101.25** Time Zone: **-9.00 (NAK)** Period: **94-19** WBAN: **25617**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
		99.6%	99%	99.6%			99%			0.4%		1%					
				DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-11.1	-8.9	-17.2	0.8	-9.9	-15.1	1.0	-8.9	18.5	-0.5	16.5	-4.1	10.0	340	0.779	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	8	3.8	17.1	13.2	15.9	12.5	14.1	11.6	14.1	15.8	13.2	14.9	12.5	13.8	3.9	160	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	13.0	9.3	14.3	12.5	9.0	13.9	12.1	8.8	13.6	39.3	15.4	37.3	14.8	35.7	13.9	18.7	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	15.3	13.0	11.5	DB	-12.3	18.8	2.2	1.4	-13.9	19.8	-15.1	20.6	-16.4	21.3	-17.9	22.3	(4)
(5)				WB	-12.9	14.5	2.2	1.5	-14.4	15.6	-15.7	16.5	-16.9	17.4	-18.4	18.5	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		5.2	0.1	0.8	0.2	2.3	5.5	8.7	11.4	12.2	10.1	6.1	3.2	1.3	(6)
(7)		DBStd		5.15	4.16	3.53	3.78	2.55	2.10	1.75	1.65	1.64	1.84	2.44	2.96	3.61	(7)
(8)		HDD10.0		1909	307	257	303	231	142	46	4	2	20	122	204	271	(8)
(9)		HDD18.3		4800	566	491	561	481	399	288	214	191	247	379	454	529	(9)
(10)		CDD10.0		151	0	0	0	0	1	8	49	69	23	1	0	0	(10)
(11)		CDD18.3		0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3		0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7		0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg		4.9	5.3	5.2	5.3	5.0	4.2	3.6	3.4	3.9	4.9	5.8	6.2	5.9	(14)
(15)	Precipitation	PrecAvg		1051	98	90	72	58	66	75	93	76	113	97	102	111	(15)
(16)		PrecMax		1430	160	185	148	95	125	125	146	113	183	166	167	239	(16)
(17)		PrecMin		789	40	29	21	19	17	32	37	28	75	49	38	41	(17)
(18)		PrecStd		155	32	37	33	21	25	23	27	21	27	31	33	43	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	7.2	7.2	8.8	13.8	16.7	18.0	18.7	16.1	12.4	9.3	7.7	(19)	
(20)			MCWB	5.4	5.6	5.3	5.6	7.8	11.7	13.6	15.1	13.1	10.7	8.1	6.7	(20)	
(21)		2%	DB	6.2	6.1	6.0	7.3	11.3	14.0	17.0	17.2	14.0	11.2	8.4	7.2	(21)	
(22)			MCWB	4.7	4.7	4.2	4.5	6.9	10.2	13.0	13.6	12.0	9.9	7.2	5.9	(22)	
(23)		5%	DB	5.1	5.2	4.9	6.3	9.1	12.7	15.2	16.0	13.1	10.2	7.7	6.3	(23)	
(24)			MCWB	3.7	3.9	3.3	4.0	6.2	9.6	11.9	13.0	11.0	8.6	6.2	4.7	(24)	
(25)		10%	DB	4.1	4.4	4.0	5.8	7.9	12.0	13.9	14.2	12.6	9.1	7.3	5.8	(25)	
(26)			MCWB	2.8	3.0	2.4	3.8	5.4	9.2	11.3	12.0	10.7	7.6	5.6	4.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	6.3	6.0	6.4	9.0	12.7	14.9	15.6	14.2	11.4	8.6	7.5	(27)	
(28)			MCDB	6.9	6.7	6.6	7.7	12.9	15.4	16.9	17.7	14.8	11.6	8.8	7.5	(28)	
(29)		2%	WB	5.1	5.2	4.6	5.2	7.8	11.2	13.7	14.3	13.0	10.1	7.7	6.4	(29)	
(30)			MCDB	6.1	6.0	5.8	6.7	10.0	13.2	15.9	16.2	13.7	10.8	8.0	7.0	(30)	
(31)		5%	WB	4.1	4.3	3.6	4.6	6.9	10.3	12.8	13.5	12.1	8.8	6.9	5.2	(31)	
(32)			MCDB	5.1	5.3	4.7	6.2	8.7	12.4	14.6	15.2	12.8	9.5	7.6	6.5	(32)	
(33)		10%	WB	3.4	3.5	3.0	3.9	6.1	9.4	12.1	12.8	11.3	8.0	5.7	4.3	(33)	
(34)			MCDB	4.4	4.5	4.0	5.4	7.9	11.4	13.4	14.3	12.4	8.8	7.2	5.7	(34)	
(35)	Mean Daily Temperature Range		MDBR	3.3	3.4	3.8	3.9	4.2	4.1	4.0	3.8	3.6	3.3	3.3	3.3	(35)	
(36)		5% DB	MCDBR	3.3	3.3	3.9	4.4	6.6	6.7	6.3	5.6	4.2	3.7	3.7	3.4	(36)	
(37)			MCWBR	3.1	3.1	3.1	3.2	4.0	4.0	3.6	3.3	2.9	3.1	3.7	3.4	(37)	
(38)		5% WB	MCDBR	3.1	3.3	3.5	4.0	5.4	5.9	5.1	4.6	3.6	3.3	3.4	3.3	(38)	
(39)			MCWBR	3.0	3.3	3.2	3.1	3.5	3.7	3.3	3.1	3.2	3.2	3.8	3.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.282	0.330	0.353	0.385	0.390	0.391	0.374	0.383	0.366	0.346	0.328	0.287	(40)	
(41)		taud		2.399	2.302	2.249	2.221	2.280	2.357	2.473	2.428	2.404	2.317	2.177	2.336	(41)	
(42)		Ebn at Noon		670	738	808	833	852	855	865	831	793	704	552	561	(42)	
(43)		Edh at Noon		49	78	103	122	122	115	100	99	89	76	60	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg		0.51	1.04	2.37	3.61	4.44	4.78	4.26	3.54	2.56	1.45	0.65	0.35	(44)	
(45)		RadStd		0.07	0.12	0.37	0.22	0.54	0.55	0.46	0.33	0.22	0.11	0.06	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	N/A	+1.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page